



Building Reversible Circuits with Quantum Gates-Hands on Session



Presented at:

 **Silicon Valley Quantum Computing Group**

 April 19, 9:30 PM IST / 9 AM PST

Dr. S. Manjula Gandhi

Associate Professor

Coimbatore Institute of Technology, India

&

IBM Qiskit Advocate

IBM Quantum Educator

Today's Agenda

- ✓ Introduction
- ✓ From Bits to Qubits
- ✓ Classical Computing vs Quantum Computing
- ✓ Quantum Computing Principles
- ✓ Quantum Computing Applications
- ✓ Reversible Circuits Vs Irreversible Circuits
- ✓ Working with IBM Qiskit Composer
- ✓ Single Qubit Quantum gates-Hands on
- ✓ Multi Qubit Quantum gates-Hands on
- ✓ Building classical logic gates using quantum circuits-Hands on
- ✓ References



INTRODUCTION

- ✓ The development of computing machines found great success in the last decades. But the ongoing **miniaturization of integrated circuits** will reach its limits in the near future.
- ✓ **Shrinking transistor sizes** and **power dissipation** are the major barriers in the development of smaller and more powerful circuits.
- ✓ **Reversible logic** provides an alternative that may overcome many of these problems in the future.
- ✓ For low-power design, reversible logic offers significant advantages since **zero power dissipation** will only be possible if computation is **reversible**.
- ✓ Furthermore, quantum computation profits from enhancements in this area, because every **quantum circuit** is inherently **reversible** and thus requires reversible descriptions.



What is a Quantum Computer??



- ✓ A quantum computer is a type of computer that uses **quantum mechanics** so that it can perform certain kinds of computation more efficiently than a regular computer can.
- ✓ Normal Computers/Classical Computers use **bits** (0/1), whereas Quantum Computers use **Quantum Bits** (Qubits).
- ✓ Each Qubit can not only be set to 1 or 0, but it can also be set 1 and 0 or anywhere between 0 and 1 at the same time.

How does the use of Qubits affect Computation?

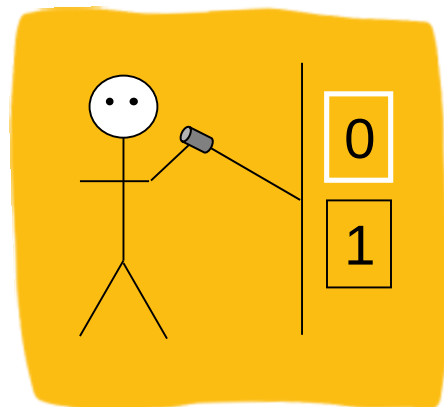
Classical Computation

Data Units : Bits

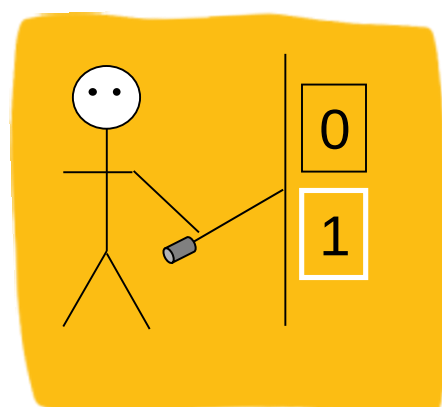
● = '1' ○ = '0'

Valid states: $x = '0' \text{ or } '1'$

$x = 0$



$x = 1$



Quantum Computation

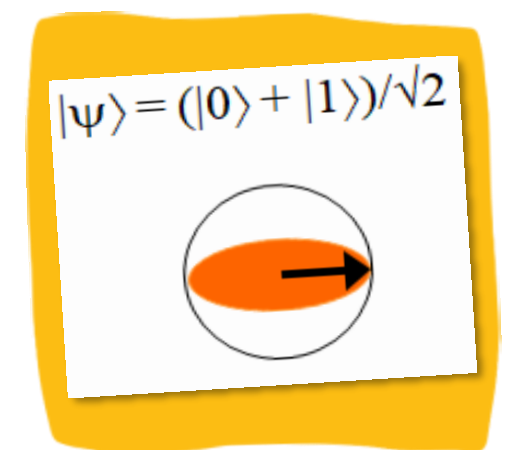
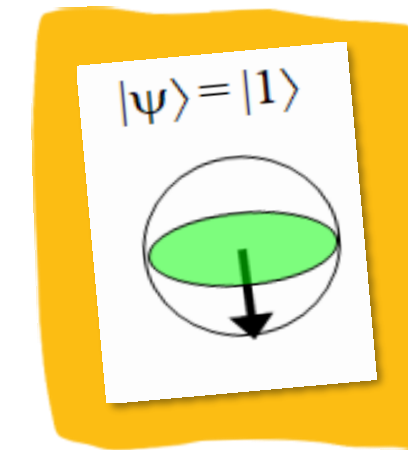
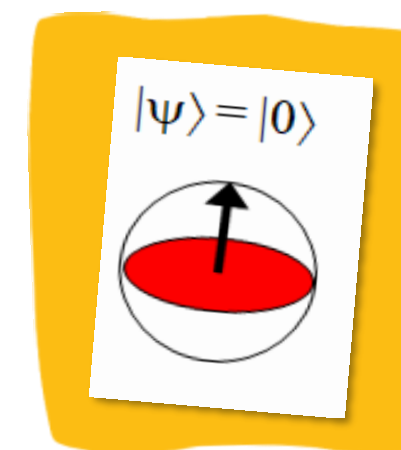
Data Units : Qubits

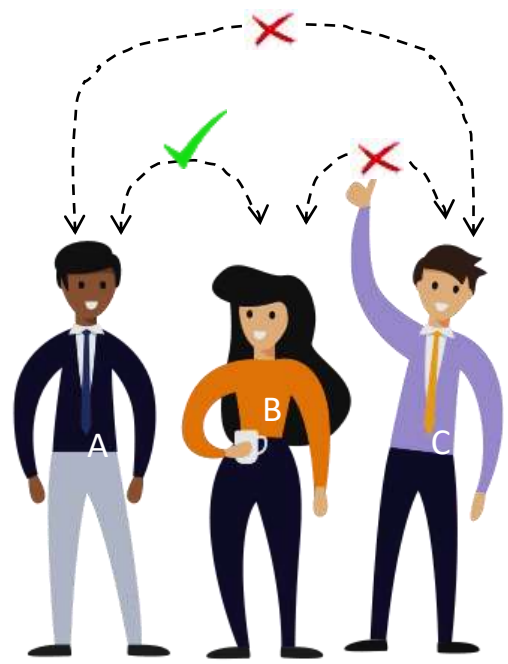
↑ = $|0\rangle$ ↓ = $|1\rangle$

Valid states:

$$|\psi\rangle = c_1|0\rangle + c_2|1\rangle$$

$$\text{where, } |c_1|^2 + |c_2|^2 = 1$$





✓ Friend Pairs
✗ Enemy Pairs

Scenario: 1

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

A	B	C	Score
0	0	0	-1
0	0	1	1
0	1	0	-1
0	1	1	-1
1	0	0	-1
1	0	1	-1
1	1	0	1
1	1	1	-1



TAXI #0



TAXI #1

Score of a given configuration
= (#friend pairs sharing the same car) - (# enemy pairs sharing the same car)

$$2^3 = 8 \text{ ways}$$





ABC Travel Agency



Scenario: 2

A	B	C	D
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

A	B	C	D	E
0	0	0	0	?
0	0	0	1	?
0	0	1	0	?
0	0	1	1	?
0	1	0	0	?
0	1	0	1	?
0	1	1	0	?
0	1	1	1	?
1	0	0	0	?
1	0	0	1	?
1	0	1	0	?
1	0	1	1	?
1	1	0	0	?
1	1	0	1	?
1	1	1	0	?
1	1	1	1	?

$2^4 = 16$ ways



Score of a given configuration
= (#friend pairs sharing the same car) - (# enemy pairs sharing the same car)



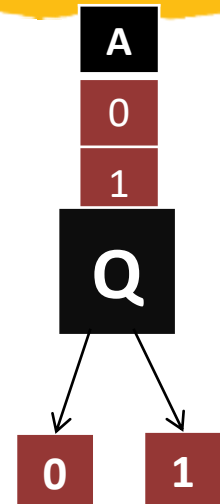
Classical Computation

- 3 people, $2^3 = 2*2*2 = 8$ ways
- 4 people, $2^4 = 2*2*2*2 = 16$ ways
- 100 people, $2^{100} = 10^{30} = \text{One million million million million million}$ configurations, which is impossible to solve with a regular computer.

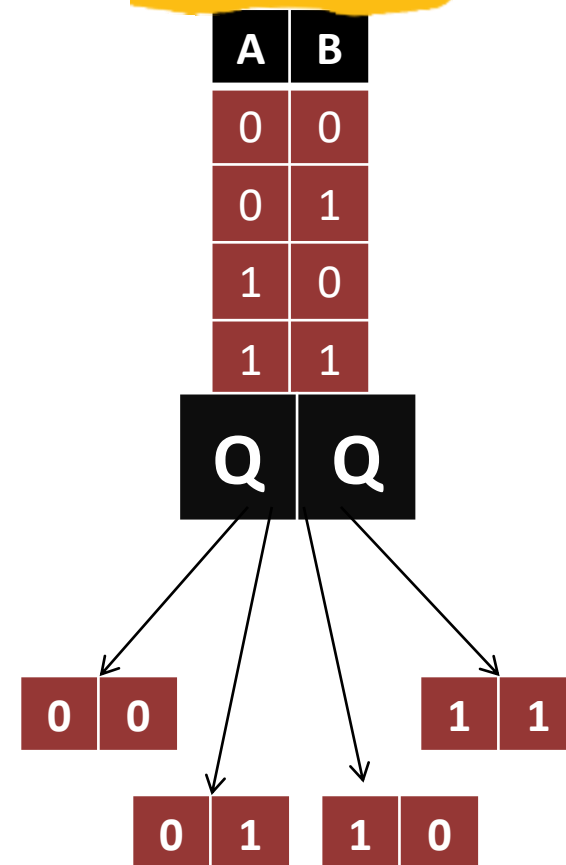


Solving this problem with a quantum computer!!

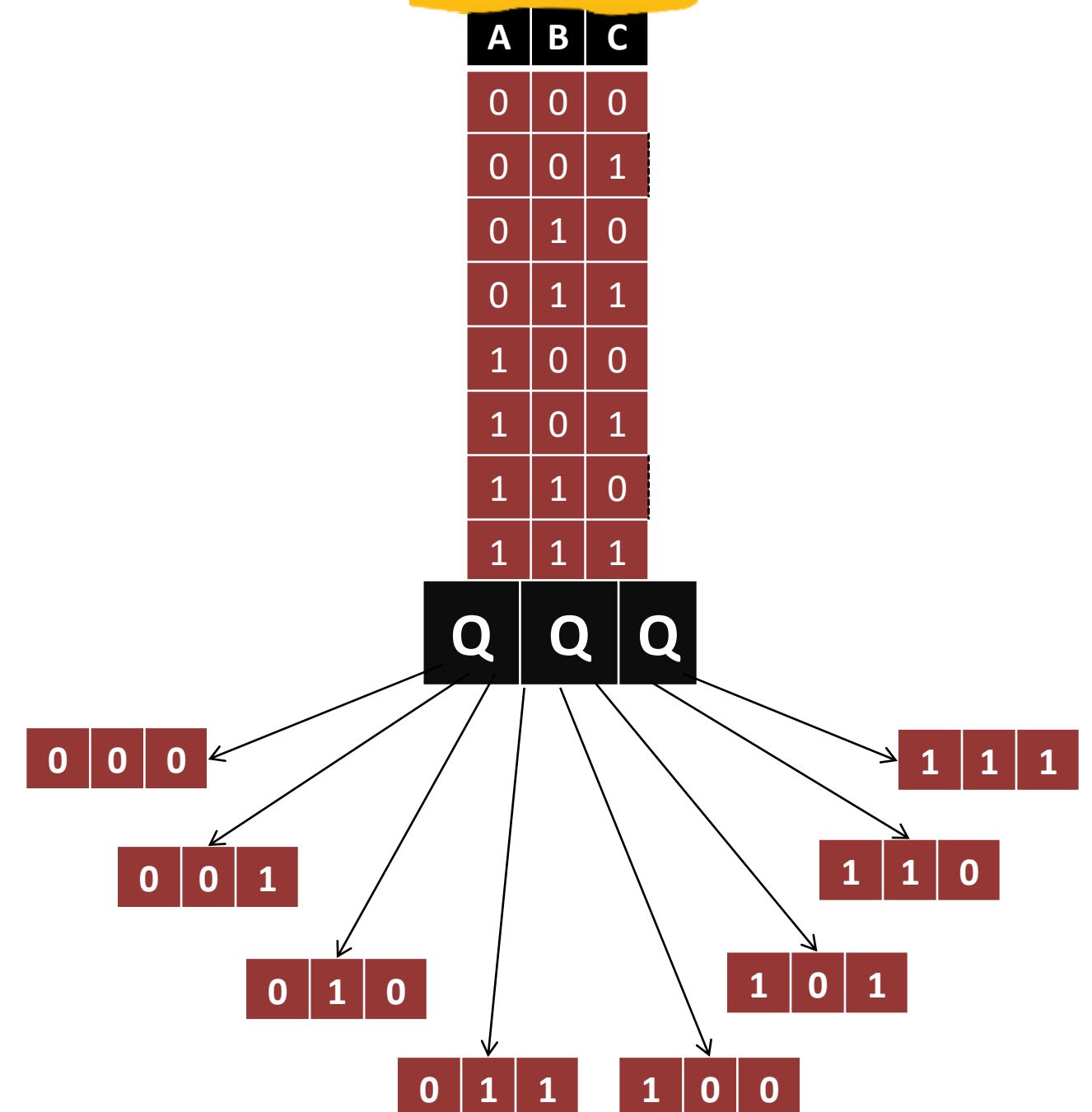
1 bit



2 bit



3 bit



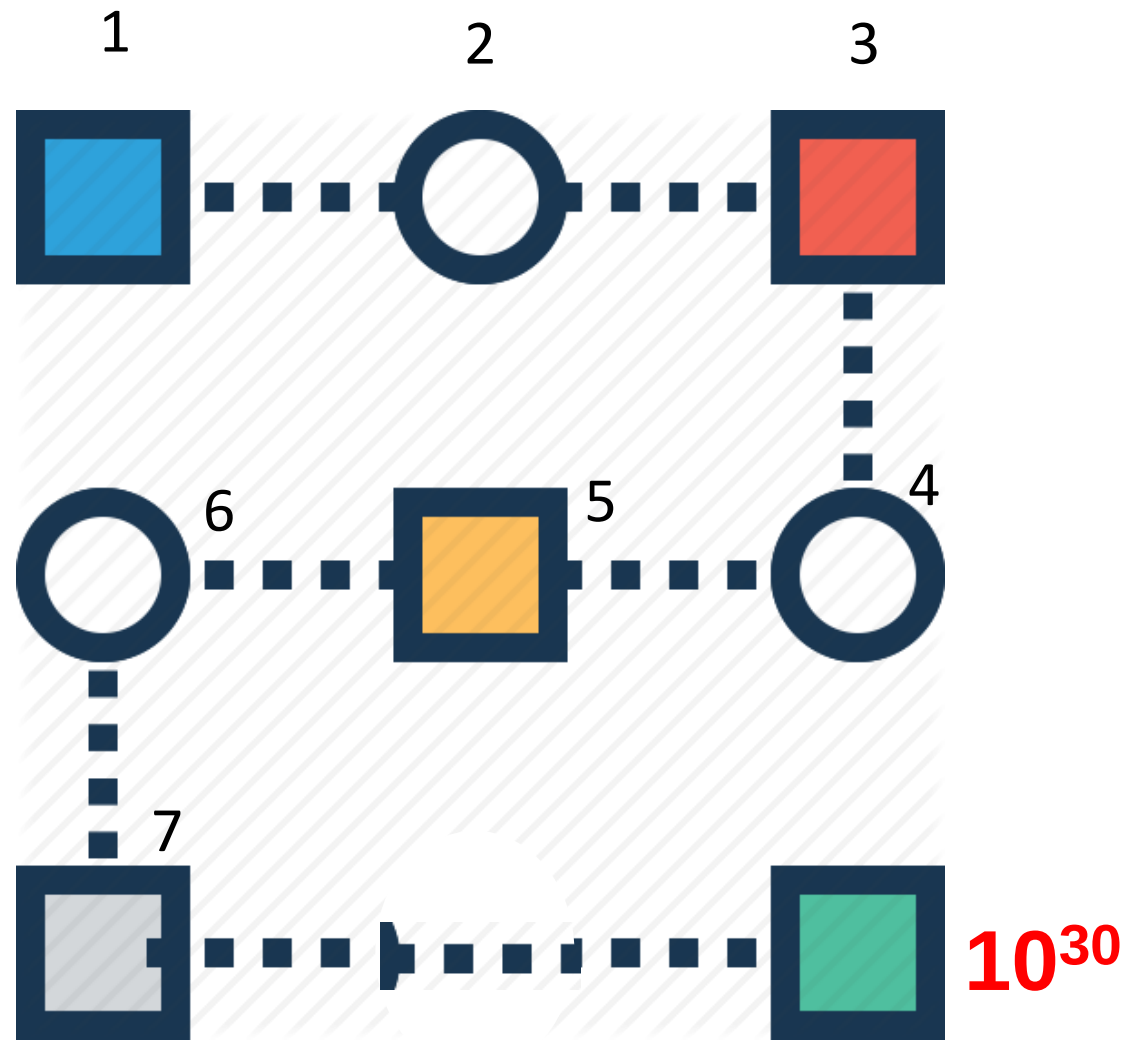


PARALLEL UNIVERSE

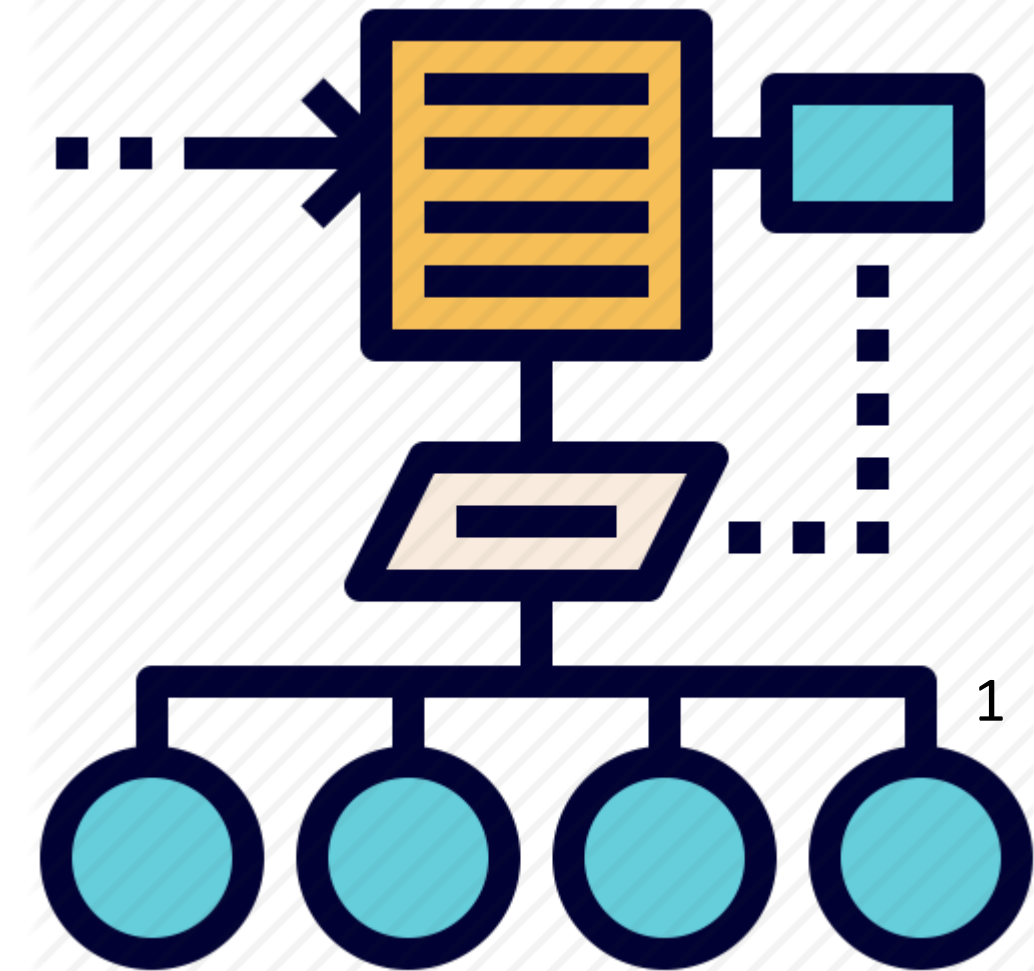
0

1

OPERATIONS



Classical Computation



Quantum Computation

DISADVANTAGES OF QUANTUM COMPUTATION



PRONE TO ERRORS

In reality, there are **errors when running a quantum computer**. So, instead of finding the best solution, it might find the second-best solution, the third best solution, and so on.



MULTIPLE ITERATIONS

You will probably want to run the same operation on a quantum computer **dozens of times or hundreds of times**. Then pick the best result out of the many results you get.



Advantages of QC

However, it's still much **better than running** the same problem on a regular computer and having to repeat the same type of computation **One million million million million million times.**



Quantum Computing Concepts



- Superposition States
- Quantum Entanglement
- Quantum Teleportation
- Quantum Algorithms
 - Deutsch's Algorithm 1985
 - Deutsch Jozsa Algorithm 1992
 - Shor's Factoring Algorithm 1994
 - Grover's Search Algorithm 1996



Quantum Computing Applications

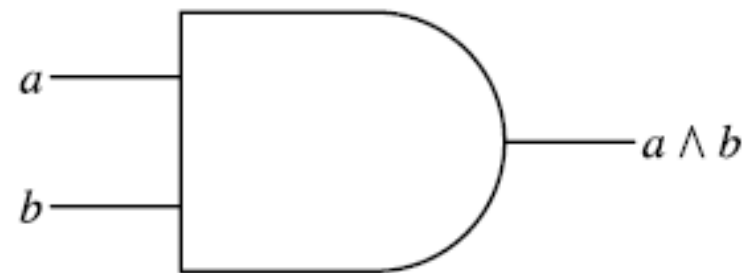
- Designing Reversible Circuits
- Quantum Software
- Quantum Hardware
- Quantum Machine Learning
- Quantum Image Processing
- Quantum Compilers
- Quantum Games
- Quantum Music
- Quantum Finance
- Quantum Optimization



Reversible Circuits/ Quantum Circuits

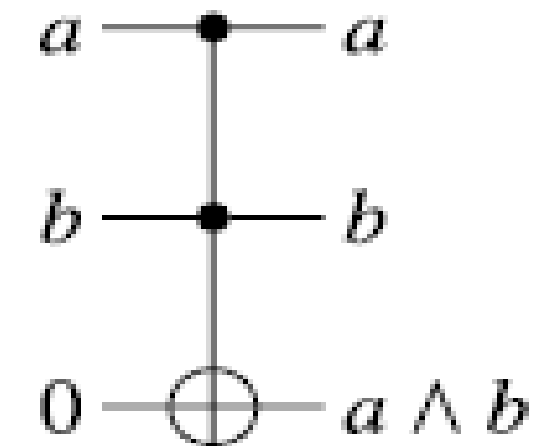
- Reversible circuits are those circuits that **do not lose information**.
- Reversible circuits can generate a **unique output vector from each input vector**, and vice versa.
- In reversible circuits, there is a **one-to-one mapping** between input and output vectors.

The Irreversible AND gate



If $a \wedge b = 0 \rightarrow A=?, B=?$

Reversible AND gate



Getting started with IBM's Qiskit

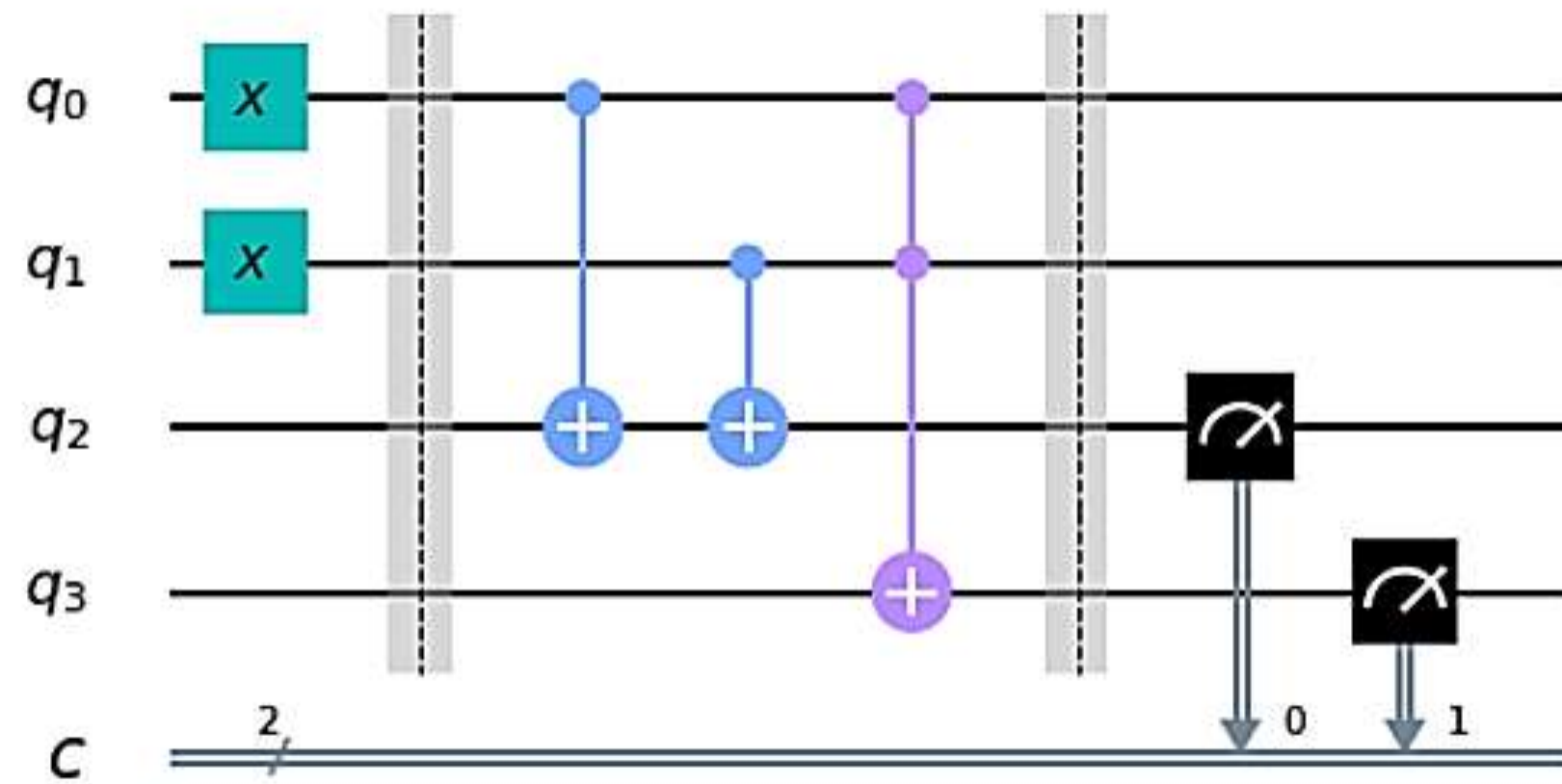


Running algorithms on quantum computers essentially involves the following steps:

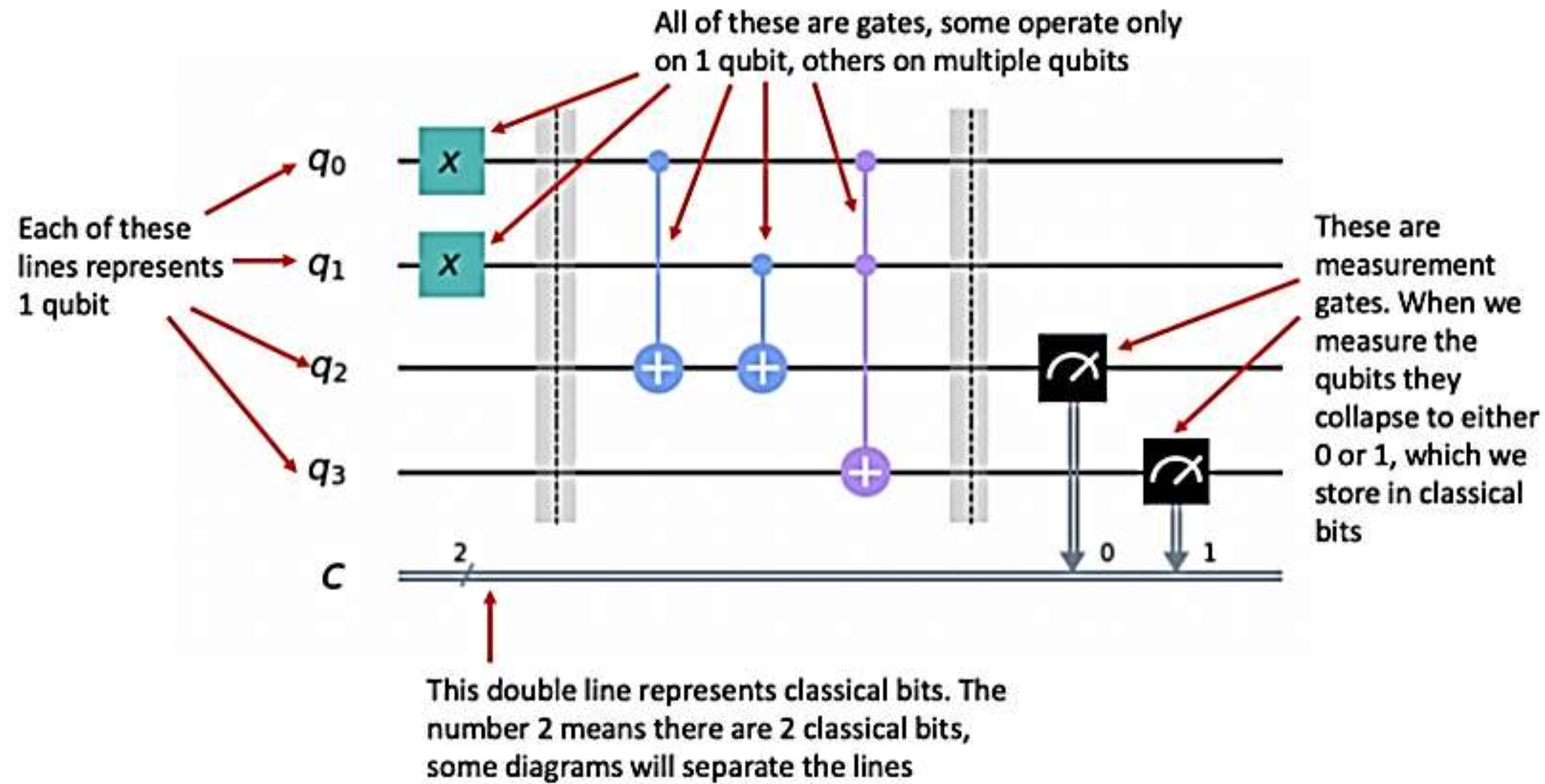
- ❖ Passing a specified number of **qubits** into the system (circuit).
- ❖ Applying a specified **combination of gates** (which make up an algorithm) to those qubits.
- ❖ **Measuring** the **qubits** and **storing** the results in **classical bits** (in 0s and 1s).



QUANTUM CIRCUIT



QUANTUM CIRCUIT



REPRESENTING QUBIT STATES

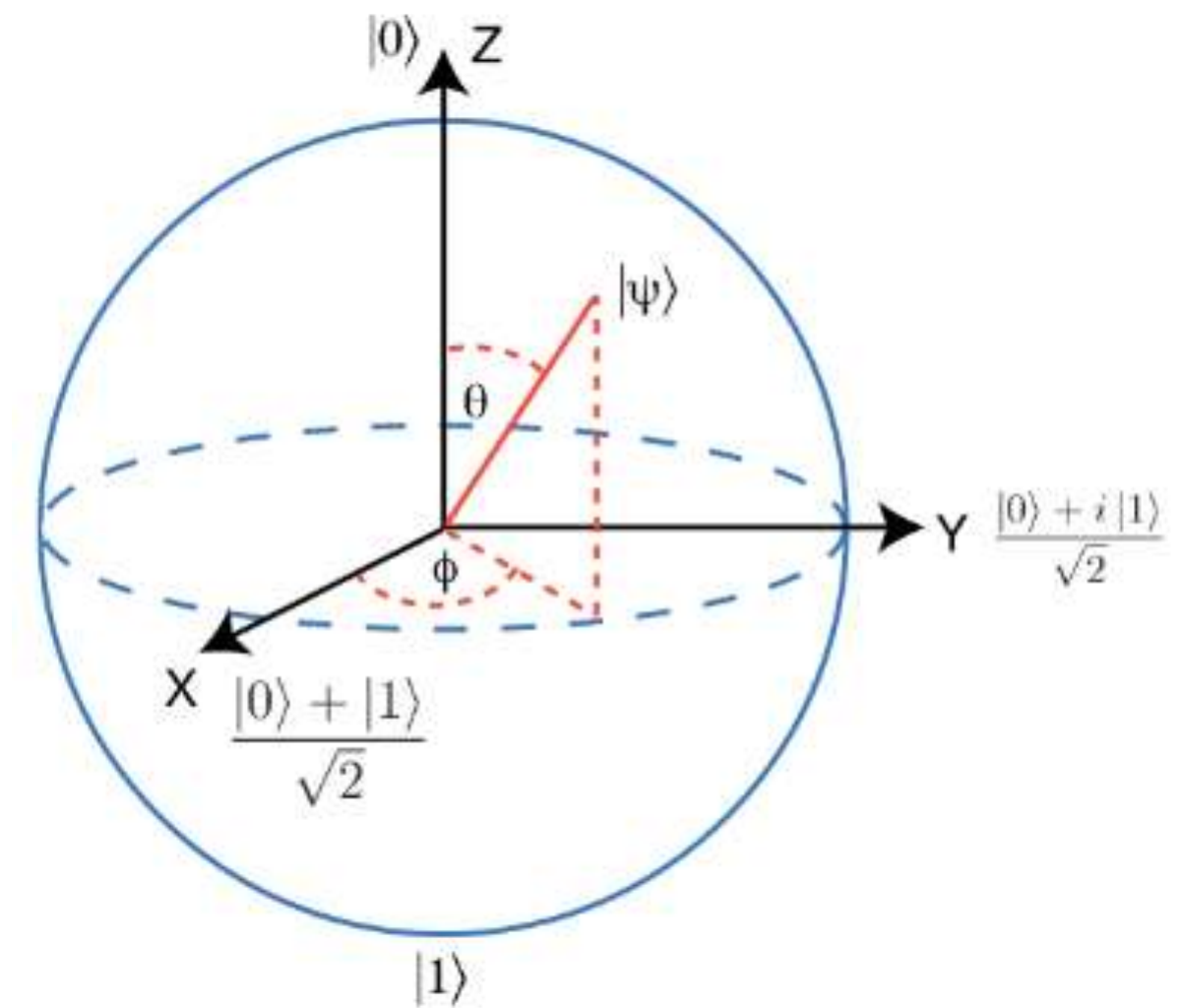
- Dirac bra-ket notation.
- Representing states using vectors.

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- Qubits could be represented by 2D vectors
- Vectors are well suited for describing quantities with a magnitude and direction

$$|q\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\frac{\theta}{2}|1\rangle$$

Where Theta and Phi are real numbers



COMPUTATIONAL BASIS

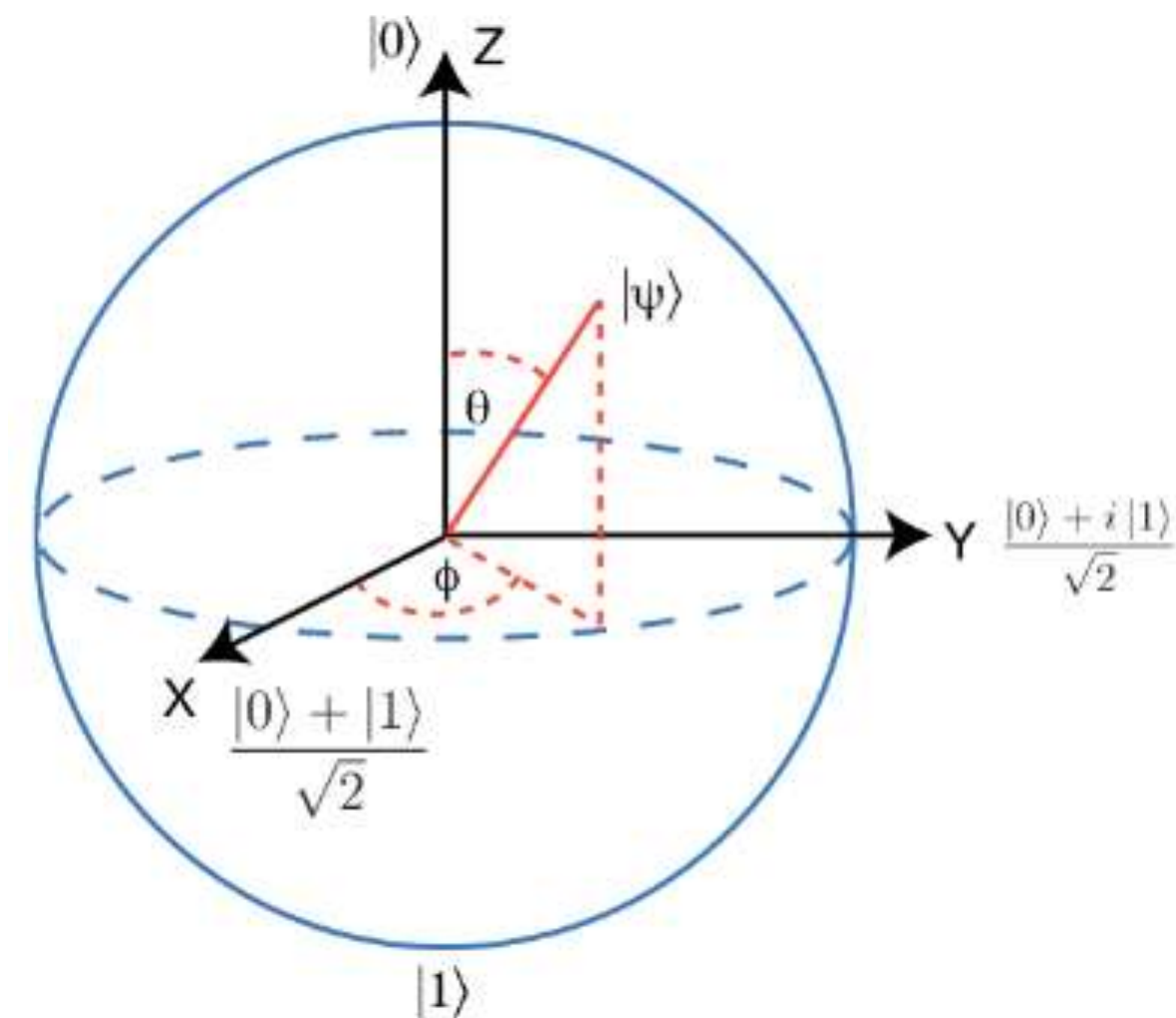
- Z basis, States $|0\rangle$ and $|1\rangle$ form an orthonormal basis, any 2D vector with a combination of these two states.
 $\{|0\rangle, |1\rangle\}$
- X basis, formed by the eigen states of the X gate, $|+\rangle$ and $|-\rangle$
 $\{|+\rangle, |-\rangle\}$

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

- Y basis, Formed by eigen states of Y gate

$$|+i\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle \quad |-i\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$

$$|q_0\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ i \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$



Single qubit gates

- `qc.x(0)` Performs a rotation of π around x axis
- `qc.y(0)` Performs a rotation of π around y axis
- `qc.z(0)` Performs a rotation of π around z axis
- `qc.s(0)` Performs a rotation of $\pi/2$ around z axis
- `qc.sdg(0)` Performs a rotation of $-\pi/2$ around z axis
- `qc.h(0)` Performs a rotation of π around an axis, located halfway between x and z axis



Single qubit gates

- X Gate(Bit Flip)

$$x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{aligned} x|0\rangle &= |1\rangle \\ x|1\rangle &= |0\rangle \\ x|+\rangle &= |+\rangle \\ x|-\rangle &= -|-\rangle \\ x|i\rangle &= i| - i\rangle \\ x| - i\rangle &= -i|i\rangle \end{aligned}$$

- Y Gate

$$y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$\begin{aligned} y|0\rangle &= i|1\rangle \\ y|1\rangle &= -i|0\rangle \\ y|+\rangle &= -i|-\rangle \\ y|-\rangle &= i|+\rangle \\ y|i\rangle &= |i\rangle \\ y| - i\rangle &= -| - i\rangle \end{aligned}$$

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$|-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$|i\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$$

$$| - i\rangle = \frac{1}{\sqrt{2}} (|0\rangle - i|1\rangle)$$

H Gate

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \begin{aligned} H|0\rangle &= |+\rangle \\ H|1\rangle &= |-\rangle \end{aligned}$$



Single qubit gates

- Z Gate (Phase Flip)

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{aligned} z|0\rangle &= |0\rangle \\ z|1\rangle &= -|1\rangle \\ z|+\rangle &= |-\rangle \\ z|-\rangle &= |+\rangle \\ z|i\rangle &= |-i\rangle \\ z|-i\rangle &= |i\rangle \end{aligned}$$

- S Gate

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

$$\begin{aligned} s|0\rangle &= |0\rangle \\ s|1\rangle &= i|1\rangle \\ s|+\rangle &= |i\rangle \\ s|-\rangle &= |-i\rangle \\ s|i\rangle &= |-\rangle \\ s|-i\rangle &= |+\rangle \end{aligned}$$



DEMO 1



- **Working with IBM Qiskit Composer**



MULTIPLE QUBIT STATES

- Single qubit has two possible states 0, 1 and a qubit state has two complex amplitudes
- Two qubits have four possible states 00, 01, 10, 11
- State of two qubits requires four complex amplitudes

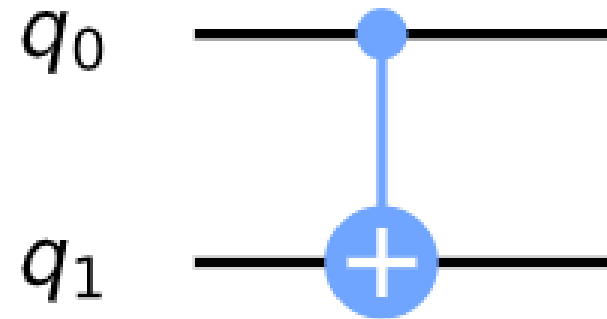
$$|a\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle = \begin{bmatrix} a_{00} \\ a_{01} \\ a_{10} \\ a_{11} \end{bmatrix}$$

$$|a_{00}|^2 + |a_{01}|^2 + |a_{10}|^2 + |a_{11}|^2 = 1$$



MULTI-QUBIT GATES

- The CNOT gate/Feynman Gate



Input (t,c)	Output (t,c)
00	00
01	11
10	10
11	01

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

```
qc = QuantumCircuit(2)
# Apply CNOT
qc.cx(0,1)
# See the circuit:
qc.draw()
```

- Operates on 2 qubits
- q0 is the 'control' qubit
- q1 is the 'target' qubit
- Used to entangle qubits



Multi qubit gates

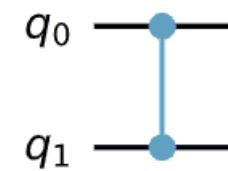
- CZ Gate

$$CZ|00\rangle = |00\rangle$$

$$CZ|01\rangle = |01\rangle$$

$$CZ|10\rangle = |10\rangle$$

$$CZ|11\rangle = -|11\rangle$$



$$CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

- The control Z gate works similar to CNOT gate , in CNOT we flip using X gate, in CZ we use Z gate.
- Z gate leaves a qubit in $|0\rangle$ untouched, while flipping the sign on a qubit in the state $|1\rangle$



Multi qubit gates

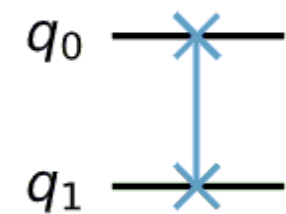
- Swap Gate

SWAP $|00\rangle = |00\rangle$

SWAP $|01\rangle = |10\rangle$

SWAP $|10\rangle = |01\rangle$

SWAP $|11\rangle = |11\rangle$



$$SWAP = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



CSWAP Gate/ FREDKIN Gate

- It is a three qubit control gate, takes three qubits as inputs.
- Uses a control qubit to determine whether or not to apply a SWAP gate to two target qubits.
- If control qubit is $|1\rangle$ then a SWAP gate is applied.
- $\text{cswap}(q_0, q_1, q_2)$

Examples:

$\text{CSWAP } |010\rangle = |010\rangle$

$\text{CSWAP } |101\rangle = |110\rangle$

$\text{CSWAP } |110\rangle = |101\rangle$

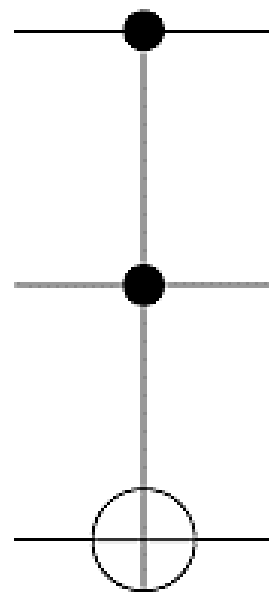
$\text{CSWAP } |111\rangle = |111\rangle$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$



CCNOT Gate/TOFFOLI Gate

- It is a three-qubit control gate, takes three qubits as inputs.
- $\text{ccx}(q0, q1, q2)$



	000	001	010	011	100	101	110	111
000	1	0	0	0	0	0	0	0
001	0	1	0	0	0	0	0	0
010	0	0	1	0	0	0	0	0
011	0	0	0	1	0	0	0	0
100	0	0	0	0	1	0	0	0
101	0	0	0	0	0	1	0	0
110	0	0	0	0	0	0	0	1
111	0	0	0	0	0	0	1	0



DEMO 2



- **Working with Multi Qubit Quantum Gates**



RECAP

- Bits vs Qubits
- Irreversible circuits vs Reversible circuits
- Creating quantum circuits using IBM Qiskit
- Visualizations – Statevector, Qsphere
- Single Qubit gates
- Multi Qubit gates



Building Reversible Circuits

- Implementing classical logic gates using quantum gates
 - Classical NOT
 - Classical AND
 - Classical NAND
 - Classical XOR
 - Classical OR
 - Classical NOR



DEMO 3



- **Constructing classical logic gates using quantum circuits**



How to become an IBM Qiskit Advocate?

- ✓ Make contributions to Qiskit code and community
- ✓ Clear IBM Certified Associate Developer-Quantum Computation using Qiskit v0.2X
<https://www.ibm.com/training/certification/C0010300>
 - Passing Qiskit Developer Certification Test is one of the requirements
- ✓ Submit application form (Applications are opened once in a year)
 - You can fill the interest form
<https://airtable.com/app8koO4BZifGFhCV/shrDgNq2U6cNIb3FQ>
(to get updates when the next round starts)

Learning Resources

1. Join Qiskit Slack

<https://airtable.com/appkU8W0NtENifBxX/shrv2nz4kGxpCAsOp>

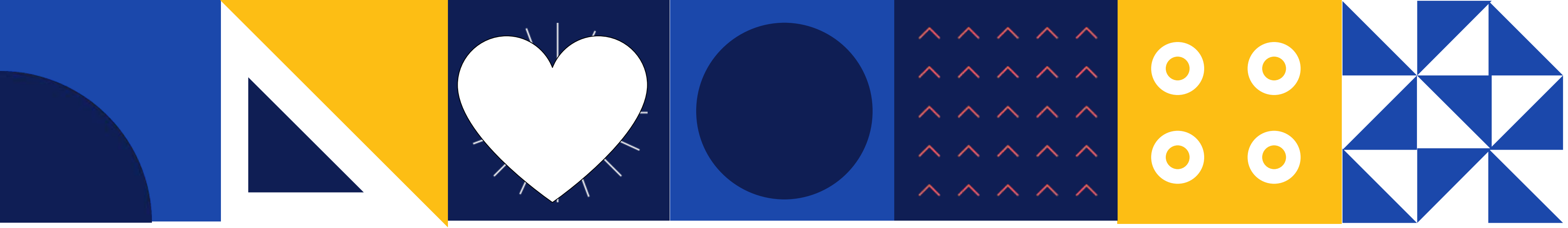
Channels- #general, #india, #quantum-internships, #textbook-reading-group, #undergraduates, #qiskit-cert-exam

2. Practice using IBM Quantum Experience

<https://quantum-computing.ibm.com/>

Benefits of becoming a Qiskit Advocate

- ✓ Advocate Mentorship Program
- ✓ Mentor – Qiskit Global Summer Schools
- ✓ Mentor – IBM Quantum Challenges
- ✓ Organizer - Qiskit Fall Fest Events
- ✓ Invited Speakers
- ✓ Advocate Café
- ✓ IBM Internships –Hardware and Software
- ✓ New Collaborations



Hey!!! Wanna Contribute to the

Future of Quantum Computing???

1. **Qiskit textbook** - <https://qiskit.org/textbook/preface.html>
2. **Qiskit Slack**-<https://ibm.co/joinqiskitslack>
3. **Participate in Hackathon's** – RELATED TO QC-IBM, MICROSOFT, Google, Xanadu, Rigetti, etc.
4. **Qiskit Documentation Localization-Hindi, Tamil, Telugu**
<https://github.com/qiskit-community/qiskit-translations>
https://qiskit.org/documentation/locale/ta_IN/index.html
5. **CIT Quantum Hackathon –Youtube videos**
<https://www.youtube.com/channel/UC5j0JSpvxYWqhvcUf880ycg>



REFERENCES

- M. A. Nielsen, I. L. Chuang, (2000) “*Quantum Computation and Quantum Information*”, Cambridge Univ. Press, Cambridge.
- Alexis De Vos, (2010) “*Reversible Computing, Fundamentals, Quantum Computing, and Applications*”, WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.
- Drechsler, R. and Wille, R., (2012) “*Reversible Circuits: Recent Accomplishments and Future Challenges for an Emerging Technology*”, *Progress in VLSI Design and Test Lecture notes in Computer Science*, 7373, 383-392.
- James L. Weaver & Frank J. Harkins, (2022) “Qiskit Pocket Guide, Quantum Development with Qiskit”, O’Reilly



REFERENCES

- Thomas G.Wong, (2022), “Introduction to Classical and Quantum Computing”, Rooted Grove
- H. Thapliyal, N. Ranganathan, “A new reversible design of BCD adder”, Design, Automation & Test in Europe, pp. 1-4, 2011.
- Hassi Norlen, “Quantum Computing in Practice with Qiskit and IBM Quantum Experience”, Packt Publishing, 2020
- Gines Carrascal, Alberto A del Barrio, Guillermo Botella, “First experiences of teaching quantum computing”, The Journal of Supercomputing, 77:2770-2799, 2021
- Robert S. Sutor, “Dancing with Qubits, How quantum computing works and how it can change the world”, Packt Publishing Ltd., 2019



REFERENCES

- Manjula Gandhi S et al, (2024) “Quantum blockchain: Trends, technologies and future directions”, IET Quantum Communication, Wiley
- Manjula Gandhi, S., Gayathri Devi, S., Sathya, K., Vani, K.H., Kiruthika, K. (March 2023), “Quantum Concepts”, In: Pandey, R., Srivastava, N., Singh, N.K., Tyagi, K. (eds) “Quantum Computing: A Shift from Bits to Qubits”, Studies in Computational Intelligence, volume 1085. Springer, Singapore. https://doi.org/10.1007/978-981-19-9530-9_6, ISBN 978-981-19-9529-3 ISBN 978-981-19-9530-9 (eBook)
- Gayathri Devi, S., Manjula Gandhi, S., Chandia, S., Boobalaragavan, P. (March 2023), “Exploring IBM Quantum Experience”, In: Pandey, R., Srivastava, N., Singh, N.K., Tyagi, K. (eds) “Quantum Computing: A Shift from Bits to Qubits”, Studies in Computational Intelligence, volume 1085. Springer, Singapore. https://doi.org/10.1007/978-981-19-9530-9_15, , ISBN 978-981-19-9529-3 ISBN 978-981-19-9530-9 (eBook)
- Adarsh Kumar, Surbhi Bhatia, Keshav Kaushik, Manjula Gandhi, Gayathri Devi, Diego Pacheco, Arwa Mashat, “Survey of Promising Technologies for Quantum Drones and Networks”, IEEE Access, September 2021, <https://ieeexplore.ieee.org/document/9527199>
- Manjula Gandhi S, Devishree J, Sathish Mohan S, “A new reversible SMG gate and its application for designing reversible two’s complement adder/subtractor with overflow detection logic for quantum computer based systems”, Springer Advances in intelligent systems and computing series, Vol. 246, 259-266, 2014.



REFERENCES

- Nati Erez , Manjula Gandhi S, Alvaro Rafael Gomez1 , Rishi Kumar Tiwari , Azuany Mila Ceron, Designing interactive introduction to quantum computing course with Classiq, September 2023 – January 2024
- Manjula Gandhi S, Hitanshu Gedam, Vardaan Sahgal, Design of Quantum Circuits for performing Arithmetic Operations, <https://github.com/qiskit-advocate/qamp-spring-23/issues/17>, March 2023-July 2023
- Manjula Gandhi S , Jayakumar Vaithiyashankar, Kavitha SS, Developing tutorials for quantum machine learning, <https://github.com/qiskit-advocate/qamp-fall-22/issues/21>, August 2022-Nov 2022
- Manjula Gandhi S , Balaji Seetharaman , Kaushal Kishor Gagan , Mitesh Nurwag , Real world data analysis using Quantum Machine Learning algorithms , <https://github.com/qiskit-advocate/qamp-spring-22/issues/18>, March 2022-June 2022
- Manjula Gandhi S, Guncha Malik, Qiskit Lab Manual, <https://github.com/qiskit-advocate/qamp-fall-21/issues/33>, October 2021-December 2021
- Julien Gacon , Manjula Gandhi S, Mantas Čepulkovskis, Arithmetic Circuit Library, <https://github.com/qiskit-advocate/qamp-spring-21/issues/12>, March 2021-May 2021



Reach me at !!!



manjulagandhi@cit.edu.in



www.linkedin.com/in/dr-manjula-gandhi-s-3a4b6a110/





THANK YOU!

Happy Learning.